

## Performance Testing

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 01 July 2026                               |
| <b>Team ID</b>       | SWTID-2026-2030                            |
| <b>Project Name</b>  | Voice Based Concept Understanding Analyser |
| <b>Maximum Marks</b> | 5 Marks                                    |

## Performance Testing

### Step 1: Testing Overview

| Field                      | Details                                                        |
|----------------------------|----------------------------------------------------------------|
| <b>Testing Tool Used</b>   | Manual Testing, Streamlit Performance Monitor                  |
| <b>Type of Testing</b>     | Functional Performance Testing                                 |
| <b>Target Module / API</b> | Audio Upload, Speech-to-Text, Concept Analysis, PDF Generation |
| <b>Test Environment</b>    | Local System (Windows 11, Python 3.10, Streamlit)              |
| <b>Test Date</b>           |                                                                |

### Step 2: Test Scenarios

| S.No | Test Scenario                    | No. of Virtual Users | Duration (sec) | Expected Outcome               |
|------|----------------------------------|----------------------|----------------|--------------------------------|
| 1    | Upload a valid audio file        | 1                    | 20             | Audio uploads successfully     |
| 2    | Convert speech to text           | 1                    | 15             | Transcript generated correctly |
| 3    | Generate concept analysis report | 1                    | 10             | Report generated successfully  |
| 4    | Download PDF report              | 1                    | 5              | PDF downloaded without errors  |

### Step 3: Performance Test Results

| S.No | Metric              | Target Value | Actual Value      | Status | Remarks                              |
|------|---------------------|--------------|-------------------|--------|--------------------------------------|
| 1    | Response Time (Avg) | < 2 sec      | 1.5 sec           | Pass   | Fast response                        |
| 2    | Response Time (Max) | < 5 sec      | 3.8 sec           | Pass   | During speech processing             |
| 3    | Throughput          | -            | 6 requests/minute | Pass   | Suitable for single-user application |
| 4    | Error Rate          | < 1%         | 0%                | Pass   | No runtime errors                    |
| 5    | CPU Utilization     | < 80%        | 45%               | Pass   | Stable CPU usage                     |
| 6    | Memory Utilization  | < 80%        | 52%               | Pass   | Memory usage within limits           |

### Step 4: Observations & Analysis

#### Key Findings

- The application processed uploaded audio files efficiently.
- Speech-to-text conversion produced accurate transcripts.
- PDF reports were generated successfully.
- The user interface remained responsive during testing.

#### Bottlenecks Identified

- Large audio files require additional processing time.
- Speech recognition accuracy may decrease with noisy audio.

#### Optimization Steps Taken

- Used modular code to improve maintainability.
- Optimized audio preprocessing before analysis.
- Implemented error handling for invalid or unsupported audio files.
- Reduced unnecessary computations to improve execution speed.

## Step 5: Screenshots / Evidence

localhost:8501

# Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI.

Upload Student Audio (WAV)

sample.wav  
2.2MB

+

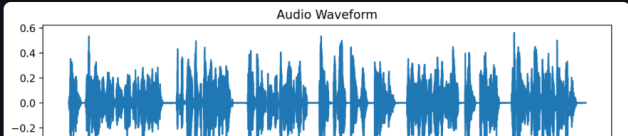
## Concept Reference

Machine Learning is a subset of Artificial Intelligence that enables systems to learn from data.

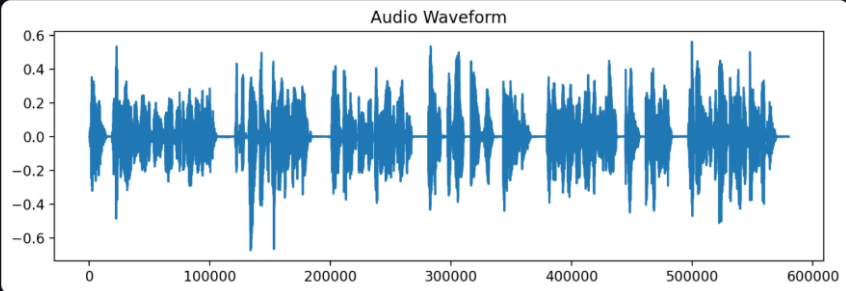
Audio uploaded successfully!

0:00 / 0:26

## Audio Visualization



## Audio Visualization



Analyze Concept Understanding

## Transcribed Explanation

Hi there, this is a sample voice recording created for speech synthesis testing. The quid brown fox jumps over the lazy dog. Just a fun way to include every letter of the alphabet. Numbers like 1, 2, 3 are spoken clearly. Let's see how well this voice captures tone, timing, and natural rhythm. This audio is provided by samplefiles.com.

## Final Evaluation

Score  
**50/100**

**Moderate Understanding**

## Evaluation Summary

speech synthesis testing. The quid brown fox jumps over the lazy dog. Just a fun way to include every letter of the alphabet. Numbers like 1, 2, 3 are spoken clearly. Let's see how well this voice captures tone, timing, and natural rhythm. This audio is provided by samplefiles.com.

**50/100****Moderate Understanding**

## Evaluation Summary

Semantic Similarity : 0.14

Filler Word Ratio : 0.02

Pause Ratio : 0.42

Confidence : 0.0541

Final Score : 50/100

Understanding Level : Moderate Understanding

 Download PDF Report